

SilentMethyl: Multimodal Prediction of CpG Methylation and Single-Nucleotide Variant–Associated Methylation Changes, with Calibrated Uncertainty and Independent Cross-Tissue Variant Validation

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Abstract

DNA methylation is a central component of epigenetic regulation, yet directly measuring how individual nucleotide variants alter local methylation remains difficult because matched genetic and epigenetic data are scarce. We present SilentMethyl, a multimodal gated-fusion framework that combines a DNABERT-2 encoder over 1,000-bp CpG-centered sequence with an MLP encoding breast epigenomic context—chromatin accessibility, six histone marks, and evolutionary conservation. Learned gates weight the two representations per locus to predict CpG methylation, and paired wild-type/mutant scoring estimates variant-associated methylation change.

Under chromosome-blocked splitting, the fusion model reached a β -value MAE of 0.0993 and ROC-AUC of 0.9680 on 26,570 held-out CpGs, exceeding both single-modality baselines, with the largest gains at CpG shores and at loci with high accessibility or H3K27ac signal.

We evaluate variant effects on an independent cohort three orders of magnitude larger than prior work: 66,495 held-out variant–CpG pairs (39,657 unique variants) from GENOA, an African American whole-blood meQTL study, harmonised to hg38 and covering 19,081 of the 26,570 held-out probes. Among genome-wide significant associations the model recovered effect direction above chance (55.3%, 95% CI 53.6–57.0) and correlated with reported effects in signed rank ($\rho = 0.152$, 0.117–0.188). Agreement declined monotonically with association strength and reached chance in the null stratum, providing an internal negative control. Discrimination of true meQTLs from tested-but-null pairs survived matching on variant–CpG distance (AUROC 0.570, 0.554–0.585), though the unmatched estimate did not exceed a distance-only baseline—a comparison we report explicitly.

Two findings generalise beyond this model. First, uncertainty evaluation on bounded bimodal targets is confounded by range compression: a zero-parameter heuristic outperforms genuine uncertainty estimators on the β scale and collapses on the unbounded logit scale, so calibration should be assessed in logit space. Second, gated multimodal fusion improves absolute methylation prediction but contributes nothing to variant-effect prediction (paired equivalence intervals bounded within ± 0.007), because context features are allele-invariant and therefore cannot create a variant effect. Together these results establish SilentMethyl as a calibrated framework linking sequence variation to local epigenetic regulation, and delimit what multimodal fusion can and cannot contribute to variant-effect prediction.

Draft status ^{NEW}

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New since the original draft (main.tex). Section 2.4 and Section 3.4 (independent variant evaluation on GENOA); Section 2.5 (block bootstrap and distance-matched negatives); Section 2.6 and Section 3.6 (uncertainty quantification); Section 3.5 (fusion/sequence equivalence). Abstract, title, introduction, discussion and limitations revised throughout.

Not yet written, in priority order. (i) regulatory enrichment against JASPAR/HOCOMOCO—the only remaining source of a novel biological finding, and the item that decides venue; (ii) multi-cohort evaluation on the 699 unpaired TCGA-BRCA tumours (data built, scoring not run); (iii) head-to-head against CpGenie and DeepCpG (repositories cloned, environments not yet stood up); (iv) repeated chromosome-blocked splits, four additional folds at seed 42—the only remaining model training; (v) conformational prediction intervals; (vi) a ClinVar figure for clinical relevance.

Bibliography. New citation keys required in `references.bib`: shang2023genoa, avsec2025alphagenome, ying2024methylogpt, gemma2012. If `references.bib` is absent at build time, this document falls back to a key-only reference list so that it still compiles.

1 Introduction

DNA methylation is a widely studied epigenetic modification involved in gene regulation, associated with genomic context, chromatin organization, and transcription [1, 2], and frequently dysregulated in disease. Single-nucleotide variants (SNVs) can influence local epigenetic states, but validating their direct regulatory effects is difficult because matched genetic and epigenetic data are limited. This motivates modeling DNA methylation to identify potentially functional variants.

Existing deep learning methods predict DNA methylation and related regulatory measures from local DNA sequence, neighboring methylation, and long-range sequence context [3–5]. While DNA sequence contains motifs and local regula-

tory patterns, sequence alone misses the epigenetic context in which they operate. Reference epigenomic data, including chromatin accessibility, histone marks, and evolutionary conservation, can supply part of this missing information. Predicting methylation at a locus and estimating the effect of a base change on methylation are also distinct tasks: a model that predicts baseline methylation accurately may still estimate variant-induced changes poorly.

Several frameworks for evaluating paired sequence changes already exist. CpGenie predicts CpG methylation and validates REF–ALT sequence changes against mQTLs [6]. INTERACT focuses on brain methylation, DeepMethylation/DDM combines sequence with reference tissue data, and Methven combines sequence with ATAC-seq accessibility [7–9].

^{REV}A specific gap motivates this work. Recent long-context variant-effect models—Enformer, Borzoi and AlphaGenome—predict many regulatory modalities at megabase scale but do not predict CpG methylation [5, 10]. Conversely, methylation foundation models such as MethylGPT operate on measured methylation profiles rather than sequence, and so cannot score an unobserved variant at all [11]. Sequence-based prediction of *variant effects on methylation* therefore sits between two active literatures without being served by either.

Existing approaches also share two methodological limitations. Most quantify the contribution of sequence and epigenomic inputs only through aggregate model-performance differences, and most do not account for reverse-complement equivariance [12]. SilentMethyl addresses both: gated fusion exposes the relative weighting of sequence and context at each individual probe, and reverse-complement stability is enforced by random orientation during training and by averaging forward and reverse-complement predictions at inference, which makes the reported output exactly RC-invariant by construction.

SilentMethyl uses synonymous mutations as a focused baseline for variant prioritization, since they change a base without changing the encoded amino acid while still potentially affecting splice recognition, RNA structure, transcript stability, translation, and exonic regulatory sequence [13, 14]. Several studies already link synonymous variants to DNA regulatory features: a synonymous *MLH1* c.27G>A allele was found in individuals with constitutional *MLH1* methylation [15]; synonymous *TP53* variants altering CpG content are linked to intragenic demethylation and alternative promoter usage [16]; synonymous *ADAMTS13* variants alter molecular properties by creating and destroying CpG sites [17].

^{REV}This work makes four contributions. (1) A gated multimodal architecture that improves CpG methylation prediction over either modality alone and exposes per-locus modality weighting. (2) Independent variant evaluation at a scale that supports stratified inference: 66,495 held-out variant–CpG pairs from an African American whole-blood cohort, evaluated against a distance-matched negative control with intervals that respect linkage disequilibrium. (3) A demonstration that uncertainty evaluation on bounded bimodal targets is con-

founded by range compression, with a scale-stability diagnostic that distinguishes genuine uncertainty signal from metric artifact. (4) A negative result with a mechanistic explanation: gated fusion cannot contribute to variant-effect prediction while its context features are allele-invariant. The last two apply to any sequence-plus-context model performing variant scoring.

2 Methods

2.1 Data construction

Methylation beta labels were constructed from the TCGA-BRCA HM450 matrix [18, 19]. Using a sample-type-11 filter, 97 solid-tissue-normal columns were selected from 893 total, verified to represent 97 unique participants without duplicate samples.

Single-probe methylation was represented as the median of patient beta values. For CpG i , if O_i is the set of patients with a recorded measurement at CpG i , then

$$\beta_i = \text{median}_{s \in O_i} \beta_{i,s}, \quad y_i = \mathbf{1}[\beta_i > 0.5],$$

$$\tilde{\beta}_i = \max\left(10^{-4}, \min\left(1 - 10^{-4}, \beta_i\right)\right),$$

$$M_i = \log_2\left(\frac{\tilde{\beta}_i}{1 - \tilde{\beta}_i}\right).$$

The beta-to- M transformation follows [20].

All probes required an exact CpG-centered DNA sequence and a valid reference context window. None of the 418,486 probes in the combined dataset were flagged by the HM450 MASK_general filter. Median probe coverage was 97 of 97, and 95.1% of loci had at least 78 valid patients. Probe coordinates and sequence were taken from the hg38 HM450 annotation resource [21, 22].

Data were split by chromosome before training: chromosomes 10–11 for validation, chromosomes 8–9 for test, remaining chromosomes for training (Table 1). Splitting by chromosome ensures no sequence windows overlap between partitions.

Each DNA input was a 1,000-bp CpG-centered sequence with the target CpG nucleotides at zero-indexed positions 499 and 500. The epigenetic input comprised seven MCF-10A attributes—ATAC-seq accessibility and six histone marks (H3K4me3, H3K27ac, H3K27me3, H3K9me3, H3K36me3, H3K4me1)—averaged across a CpG-centered 100-bp window, plus two PhyloP scores for the two nucleotides of the central CpG. A missingness mask accompanied each feature; missing values were imputed with the feature’s global median. Reference data were derived from shared resources and were not patient-specific.

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Figure 1: SilentMethyl data construction, model architecture, and evaluation workflow. Panel A shows methylation target construction, chromosome partitioning, and model inputs. Panel B shows FWD-RC orientation and staged training. Panel C shows the gated multimodal architecture. Panel D shows paired WT-MUT scoring, held-out probe testing, external mQTL validation, and synonymous-variant ranking.

Table 1: Chromosome split dataset. Methylated means median $\beta > 0.5$.

Split	Loci	Mean β	Methylated
Train	345,359	0.4948	52.25%
Validation	46,557	0.5035	53.19%
Test	26,570	0.5029	53.20%

2.2 Models, training, and evaluation

The sequence module comprised a pretrained DNABERT-2 (117M) transformer [23] followed by a convolutional layer and masked attention pooling:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{u \in V} \exp(s_u)}, \quad \mathbf{h}_{\text{seq}} = \sum_{t \in V} \alpha_t \mathbf{e}_t,$$

where V contains non-padding positions and $\mathbf{e}_t$ is the embedding at position t , producing a 768-dimensional $\mathbf{h}_{\text{seq}}$.

The context module was an MLP mapping the 18-dimensional epigenetic input (9 features + 9 missingness masks) to a 768-dimensional $\mathbf{h}_{\text{ctx}}$.

After LayerNorm produced $\tilde{\mathbf{h}}_{\text{seq}}$ and $\tilde{\mathbf{h}}_{\text{ctx}}$, the two were concatenated and passed through a gating MLP yielding scalars g_{seq} and g_{ctx} . The fused embedding was

$$\mathbf{z} = g_{\text{seq}} \tilde{\mathbf{h}}_{\text{seq}} + g_{\text{ctx}} \tilde{\mathbf{h}}_{\text{ctx}},$$

fed to two prediction heads. The regression head predicted M , converted by $\hat{\beta} = 2^{\hat{M}} / (1 + 2^{\hat{M}})$; the classification head predicted $\hat{y} = \mathbf{1}[\sigma(\text{logit}) > 0.5]$. The objective was Huber loss ($\delta = 1.345$) on regression plus binary cross-entropy on classification.

Sequence and context models were trained independently with AdamW, with DNABERT-2 fully fine-tuned. Sequence, context, and fusion training used up to 10, 15, and 10 epochs at learning rates 2×10^{-5} , 10^{-3} , and 5×10^{-4} . During fusion training the component encoders were frozen and only normalization, gating, and prediction modules were updated.

During training, forward or reverse-complement orientation was chosen with equal probability per probe. For seeds 42, 43, and 44 the checkpoint was selected by lowest validation beta MAE. Primary metrics were RMSE and MAE for β and M , and ROC-AUC for classification.

2.3 Paired sequence scoring and sSNV construction

For each SNV the predicted methylation difference was averaged across strands:

$$\Delta\hat{\beta} = \frac{(\hat{\beta}_{\text{MUT,FWD}} - \hat{\beta}_{\text{WT,FWD}}) + (\hat{\beta}_{\text{MUT,RC}} - \hat{\beta}_{\text{WT,RC}})}{2},$$

with $\Delta\hat{M}$ defined analogously. Because the reported prediction is $\frac{1}{2}[f(x) + f(\text{RC}(x))]$, it is exactly invariant under reverse complementation.

Tagged somatic sSNVs were obtained through the GDC with amino-acid preservation verified against hg38 and GENCODE v44 [19, 24]. Variants were matched to the nearest HM450 probe; 440 unique variants remained after removing MASK_general1-flagged probes and variants outside the 1,000-bp window. Variants were prioritized by $|\Delta\hat{\beta}|$ from the three-seed fusion ensemble.

2.4 Independent variant evaluation: GENOA ^{NEW}

External validation used cis-meQTL summary statistics from GENOA, an African American whole-blood cohort [25]. Three properties make it suitable: it is a different tissue, a different ancestry, and a different array generation from the training data, and it is openly released.

Harmonisation. Summary statistics are reported in hg19. The CpG was never lifted over: array probe identifiers are platform-stable, so the reported probe joins directly to the

hg38 manifest and inherits hg38 coordinates. Only SNP positions required hg19→hg38 conversion. Of 942,186 candidate pairs, 731 (0.1%) had neither reported allele matching the hg38 reference base and were discarded; this rate is itself evidence that the liftover and strand convention are correct.

Effect-allele alignment. GENOA effect sizes are GEMMA output, in which β is the effect per copy of allele1, the minor allele [26]. The model’s prediction is the effect of substituting ALT for the hg38 reference. These directions coincide only when ALT is the minor allele. In 106,628 pairs (11.3%) the hg38 reference base is the minor allele, so ALT is the major allele and the two conventions are opposed. All comparisons therefore use the published effect re-signed to the REF→ALT direction. This correction is silent if omitted—it does not produce an error—and it attenuates both signed correlation and direction agreement.

Strata. Pairs were labelled by the split membership of their target probe, read from the actual training partitions rather than inferred from chromosome number. Only the held-out stratum supports an independent-validation claim: 66,495 pairs, 39,657 unique variants, 19,081 of the 26,570 held-out probes. Pairs whose variant falls at the target CpG itself were excluded (the architecture is defined around a CpG at window indices 499:501, and such array measurements are a known SNP-under-probe artifact), as were pairs outside the 1,000-bp window, for which the model returns a delta of exactly zero. Variants that create or destroy a CpG dinucleotide have a near-deterministic direction requiring no model and are reported as a separate stratum throughout; unless stated otherwise, results below refer to the 42,866 non-CpG-altering held-out pairs.

2.5 Statistical framework ^{NEW}

Dependence. The held-out pairs comprise 39,657 variants in linkage disequilibrium across 19,081 probes and are far from independent. All intervals are percentile bootstrap intervals resampling whole 1-Mb genomic blocks rather than individual pairs; row-level resampling would treat correlated observations as independent and yield intervals several times too narrow.

Distance matching. Significant meQTLs lie closer to their target CpG than tested-but-null pairs (median 191 versus 258 bp), and predicted effect magnitude increases with proximity ($\rho = -0.29$ between $|\Delta M|$ and distance). Variant-CpG distance alone therefore separates the two classes, and an unmatched discrimination estimate partly measures distance rather than regulatory effect. Each significant pair was matched to a null pair at the same base-pair distance, sampled without replacement so that no null could be reused. Within the matched cohort, distance alone achieves AUROC 0.500 by construction, so residual discrimination is attributable to the model. The distance-only baseline is reported alongside every discrimination estimate.

2.6 Uncertainty quantification ^{NEW}

Averaging forward and reverse-complement predictions makes the output invariant but discards the disagreement between them. That disagreement is available at every locus at no additional cost and was evaluated as a single-model uncertainty estimate against a three-seed ensemble standard deviation, a zero-parameter boundary heuristic $-\widehat{\beta} - 0.5$, and a random control, using risk-coverage curves and area under the risk-coverage curve.

Because β is bounded on $[0, 1]$ and bimodal, absolute β error is mechanically compressed near the boundaries. Every comparison was therefore repeated on the unbounded M scale, and additionally within strata of predicted β at 10, 20, and 50 levels, so that an estimator’s apparent skill could be separated from headroom in the target.

2.7 External eGTEx mQTL analysis

Significant eGTEx Breast Mammary Tissue lead mQTLs ($q < 0.05$) were overlapped with test-set CpGs [27]. After excluding variants outside the sequence window or altering their target CpG, 81 associations with 70 unique variants remained. Predicted ALT-REF changes were compared with reported allele slopes using signed and absolute Spearman correlation, sign agreement, and AUROC. A sensitivity analysis dropped mutations within or immediately adjacent to a probe. A matched analysis tested whether predicted magnitude separated significant leads from high- q associations ($q \geq 0.5$) matched on chromosome and minor allele frequency (35 pairs from 53 significant and 51 high- q variants).

3 Results

3.1 Multimodal fusion outperforms single-modality models

On the 26,570-probe test set the fusion model was strongest across all metrics and seeds. The context MLP established a baseline beta MAE of 0.1395 ± 0.0001 and ROC-AUC 0.9187 ± 0.0001 . The sequence model reduced beta MAE by 21.2% (0.1099 ± 0.0008) and raised ROC-AUC to 0.9569 ± 0.0007 . Fusion improved beta MAE by a further 9.6% (0.0993 ± 0.0025) and raised ROC-AUC to 0.9680 ± 0.0021 (Fig. 2). Predictions were stable at lower coverage: on probes measured in at least 78 of 97 patients the seed-42 fusion model reached beta MAE 0.09669 and ROC-AUC 0.97066.

Improvement was consistent across seeds and across genomic regions. Fusion improved sequence-only MAE by 0.01043 (95% CI 0.00951–0.01132) and ROC-AUC by 0.01053 (0.00909–0.01207) under a 257-block resampling design.

Figure model_incremental_performance.png
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Figure 2: Held-out chromosome performance. Gray lines connect the same training seed. (A) M-value MAE. (B) beta MAE. (C) ROC-AUC.

Table 2: Held-out chr8–9 performance. Mean $\pm$ SD across three seeds on 26,570 CpGs.

Metric	Context	Sequence	Fusion
M-value MAE	1.4574 $\pm$ 0.0023	1.1941 $\pm$ 0.0070	1.0971 $\pm$ 0.0192
Beta MAE	0.1395 $\pm$ 0.0001	0.1099 $\pm$ 0.0008	0.0993 $\pm$ 0.0025
ROC-AUC	0.9187 $\pm$ 0.0001	0.9569 $\pm$ 0.0007	0.9680 $\pm$ 0.0021

3.2 Reference context contributes across regulatory conditions

Fusion reduced beta MAE in every stratum of GENCODE region, CpG island status, ATAC-seq quartile, and H3K27ac quartile. By GENCODE annotation the reduction ranged from 0.0088 at intergenic probes to 0.0130 at UTRs. By island status the largest gain was at shores (0.0162). Fusion lowered beta MAE by 0.0139 in the highest ATAC-seq quartile and 0.0147 in the highest H3K27ac quartile (Fig. 3). Context therefore contributes most where regulatory state is expected to carry additional information.

3.3 Breast mQTLs support predicted effect direction

On 81 independent breast mQTL associations the three-seed fusion ensemble recovered 64 of 81 effect directions (79.0%), achieved signed Spearman $\rho = 0.610$, and separated positive from negative slopes with AUROC 0.872. The pattern held after removing variants on or adjacent to a probe (53 associations; $\rho = 0.610$, 75.5% direction agreement, AUROC 0.872).

Absolute magnitudes agreed far less well ($\rho = 0.217$ overall, 0.282 after filtering), and the model did not separate matched significant leads from high- q leads on $|\Delta\hat{M}|$ (AUROC 0.512, 95% CI 0.380–0.646). At $n = 81$ these estimates are imprecise, which motivated the larger independent evaluation below.

Figure fusion_gain_by_epigenomic_context.png
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Figure 3: Fusion gain across epigenomic contexts. Intervals re-sample 1-Mb regions within each stratum. (A) CpG island status. (B) ATAC-seq quartile. (C) H3K27ac quartile.

Table 3: Independent variant evaluation on GENOA. Held-out stratum, non-CpG-altering pairs, three-seed ensemble. Brackets are 95% intervals from a block bootstrap over 1-Mb blocks. Signed ρ and direction agreement are computed on genome-wide significant pairs ($p < 5 \times 10^{-8}$, $n = 4,037$); AUROC uses all 42,866 pairs.

Metric	Fusion	Sequence
Signed ρ	+0.152 [+0.117, +0.188]	+0.150 [+0.109, +0.184]
Direction agreement	0.553 [0.536, 0.570]	0.554 [0.538, 0.573]
AUROC, distance-matched	0.570 [0.554, 0.585]	0.562 [0.546, 0.578]
AUROC, unmatched	0.600 [0.586, 0.614]	0.603 [0.589, 0.617]
AUROC, distance only	0.595 [0.583, 0.609]	

3.4 Independent variant evaluation at scale ^{NEW}

Scoring 66,495 held-out GENOA pairs increased the independent variant evaluation from 81 associations to 39,657 unique variants, a scale that supports stratified inference rather than a single pooled estimate (Table 3).

Effect direction was recovered above chance and signed rank correlated with reported effects, with both intervals excluding the null. Effect sizes are modest: $\rho \approx 0.15$ and 55% direction agreement describe a weak predictor, and we report it as such. What the analysis establishes is that the effect is real and measurable on an independent cohort differing in tissue, ancestry, and platform generation, on probes the model never saw.

An internal negative control. GENOA publishes every cis pair tested rather than only the meQTLs discovered, and 71% of held-out non-CpG-altering pairs have $p > 0.05$. Stratifying by association strength shows agreement rising with the strength of the underlying association and reaching chance where no association exists (Table 4, Fig. 4). The null stratum is statistically indistinguishable from zero on both metrics, providing a negative control internal to the same dataset. Intermediate strata are not strictly monotone and their intervals overlap; the interpretable contrast is significant versus null.

Table 4: Agreement by association strength. Fusion, three-seed ensemble, non-CpG-altering held-out pairs.

Stratum	n	Signed ρ	Direction
$p < 5 \times 10^{-8}$	4,037	+0.152 [+0.117, +0.184]	0.553
0.05–0.5	15,891	+0.023 [+0.006, +0.040]	0.508
$p > 0.5$	13,330	−0.004 [−0.020, +0.014]	0.497

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Figure 4: Agreement with GENOA rises with association strength and reaches chance where associations are null. Signed Spearman correlation (left) and direction agreement (right) across seven strata of reported p -value, with 95% block-bootstrap intervals.

Figure discrimination_vs_distance.png
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Figure 5: Discrimination survives distance matching. AUROC for separating genome-wide significant meQTLs from tested-but-null pairs, unmatched and within distance strata, against the distance-only baseline.

Discrimination is not explained by distance alone. Variant–CpG distance by itself separates true meQTLs from null pairs at AUROC 0.595 (0.583–0.609), statistically indistinguishable from the model’s unmatched 0.600 (0.586–0.614). We state this explicitly: *marginally, predicted effect magnitude does not outperform distance*. After matching each significant pair to a null pair at identical distance, however, discrimination remains at 0.570 (0.554–0.585), and exceeds chance in every distance bin examined. The model’s signal is therefore largely orthogonal to distance rather than a restatement of it (Fig. 5).

3.5 Gated fusion does not contribute to variant-effect prediction ^{NEW}

Fusion and sequence-only models were statistically equivalent on every variant-effect metric. Paired differences, bootstrapped over the same genomic blocks, were +0.0026 (−0.0020, +0.0070) for signed ρ , −0.0007 (−0.0064, +0.0051) for direction agreement, and −0.0021 (−0.0053, +0.0011) for within-distance-bin AUROC. The intervals are narrow enough to support equivalence rather than merely a failure to detect a difference: any difference is bounded within ± 0.007 .

This follows from the architecture. The context vector depends on the probe, not on the allele, and is therefore identical for wild-type and mutant input. Writing the fused embedding difference,

$$\Delta \mathbf{z} = [\tilde{\mathbf{h}}_{\text{seq}}^{\text{mut}} g_{\text{seq}}^{\text{mut}} - \tilde{\mathbf{h}}_{\text{seq}}^{\text{wt}} g_{\text{seq}}^{\text{wt}}] + \tilde{\mathbf{h}}_{\text{ctx}} [g_{\text{ctx}}^{\text{mut}} - g_{\text{ctx}}^{\text{wt}}],$$

the context term survives only through a gate shift that is itself driven by the sequence change. Context can modulate a variant effect; it cannot create one.

The gate is the only route by which allele-invariant context could still shape a prediction, and that route is also closed: the fusion-minus-sequence difference is flat across quartiles of gate DNA share, with all eight intervals covering zero and the largest point estimate 0.007. A faint sign pattern runs in the predicted direction across quartiles but is not statistically supported.

This is consistent with the earlier observation that fusion and sequence rankings of the 440 synonymous candidates were highly concordant (signed $\rho = 0.981$, 96.6% sign agreement). The present analysis explains why, and bounds the effect.

3.6 Uncertainty is quantifiable, but the standard metric is confounded ^{NEW}

Forward/reverse-complement disagreement tracked per-locus error at Spearman $\rho \approx 0.50$ for the sequence and fusion models (0.35–0.39 for context-only), consistently across seeds. On selective prediction, however, the zero-parameter heuristic $-\lceil \beta - 0.5 \rceil$ outperformed every genuine estimator—including the three-seed ensemble—in 8 of 9 model–seed combinations.

That result is an artifact of the target rather than a property of the estimators. Absolute β error is mechanically compressed near 0 and 1, so a model that is merely confident at extreme predictions appears well calibrated. Repeating the comparison on the unbounded M scale, and stratifying finely by predicted β , separates the two explanations: the heuristic’s within-stratum correlation collapses from +0.0369 to +0.0099 between the β and M scales, while cross-seed ensemble standard deviation (+0.0915 to +0.0952) and FWD–RC disagreement (+0.0678 to +0.0793) are unchanged (Fig. 6). At 50 strata both genuine estimators outperform the heuristic on both scales.

Scale-instability is the signature of a metric artifact. A genuine uncertainty signal ranks error about equally well on

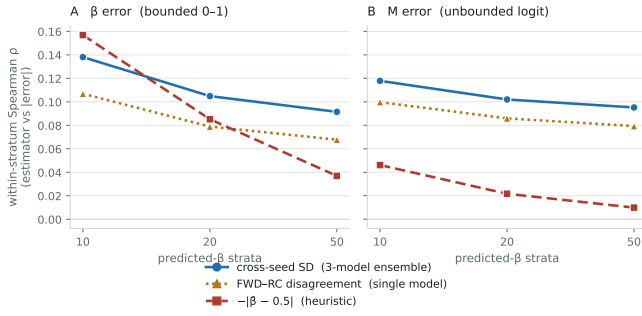


Figure 6: Scale stability separates uncertainty signal from metric artifact. Within-stratum rank correlation between each uncertainty estimator and absolute error, on the bounded β scale and the unbounded M scale. Genuine estimators are scale-stable; the boundary heuristic is not.

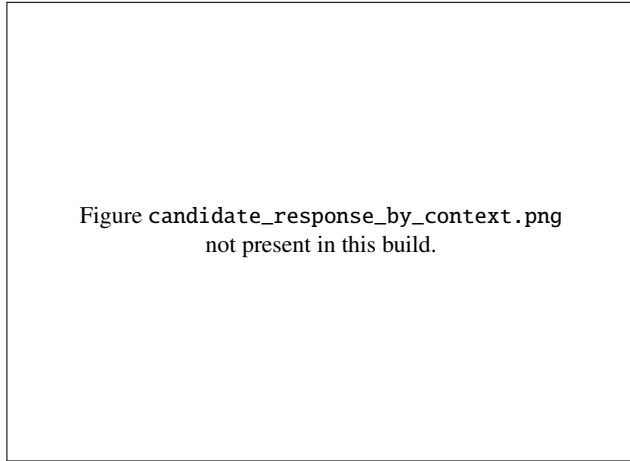


Figure 7: Candidate response by biological context. Fusion difference magnitude grouped by (A) variant region and (B) distance to CpG.

either scale; a compression artifact does not. We recommend that calibration for bounded bimodal targets be assessed in logit space, and note that β MAE is the field’s routine reporting choice.

FWD-RC disagreement retains 75–83% of the ensemble’s incremental information depending on the measure, from a single model at no additional inference cost.

3.7 Variant prioritization and case studies

Applied as a prioritization engine to 440 synonymous variant–CpG pairs, the fusion model produced consistent rankings across seeds ($r = 0.681\text{--}0.699$). Median $|\Delta\beta|$ was 0.00360 at promoter/TSS variants, 0.00261 at UTRs, and 0.00165 in gene bodies; variants within 50 bp of the target CpG had median 0.00512 against 0.00111 at 251–500 bp (Fig. 7). Neither genomic region nor distance was supplied as a feature.

The highest-ranked synonymous candidate, chr8:g.70148394G>A in *NCOA2*, is a GGC>GGT substitution 9 bp from probe cg20699548, which lies in an OpenSea context in the top quartile of both ATAC-seq and H3K27ac signal. The ensemble predicted a hypomethylating shift of

$\Delta\hat{\beta} = -0.1798 \pm 0.0198$, exceeding all 66 matched synonymous background variants (Fig. 8). *NCOA2* encodes a nuclear receptor coactivator with reported involvement in breast tumour growth [28]; no published evidence links this specific variant to *NCOA2* activation, and the prediction constitutes a testable hypothesis rather than a finding.

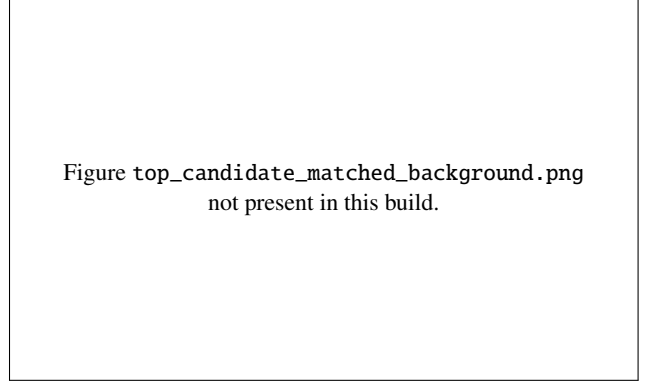


Figure 8: Top candidate versus matched background. The absolute response of the *NCOA2*-associated candidate compared with 66 matched observed synonymous variants.

The framework extends to any single-nucleotide change. Among studied nonsynonymous variants in breast-cancer genes, the two highest-ranked were both in *STK11* [29]: rs2145420809 ($\Delta\hat{\beta} = 0.0774$ at cg16601904, immediately adjacent) and rs148928808 ($\Delta\hat{\beta} = -0.0613$ at cg08681293, 204 bp downstream), the latter a known pathogenic truncating variant. Both target CpGs lie on training chromosomes, so this is an illustrative application rather than validation.

4 Discussion

SilentMethyl unifies two tasks in regulatory genomics: predicting baseline CpG methylation and estimating how that state responds to single-nucleotide variation. Integrating local sequence with breast epigenomic context consistently improved methylation prediction over either modality alone, with the largest gains at CpG shores and at loci with high accessibility or H3K27ac signal—regions where methylation is expected to reflect an interaction between sequence and chromatin state rather than sequence alone [1, 2, 30].

^{REV}The independent variant evaluation is the central addition of this revision, and it supports a specific and bounded claim. On 39,657 variants from a cohort differing in tissue, ancestry and platform generation, predicted effects recovered direction above chance and correlated with reported effects in signed rank, and discrimination of true meQTLs from tested-but-null pairs survived matching on variant–CpG distance. The effect sizes are modest and we do not present them otherwise. Their value lies in three properties the earlier 81-association analysis could not provide: the scale to stratify rather than pool, an internal negative control in the form of a null stratum that returns to chance, and intervals that respect linkage disequilibrium.

The distance analysis deserves emphasis because it is the most likely source of an inflated claim in this literature. Predicted effect magnitude is strongly related to variant–CpG proximity, true meQTLs are closer to their CpGs than null pairs, and distance alone therefore reaches AUROC 0.595—indistinguishable from the model’s unmatched estimate. Reporting the unmatched number in isolation would substantially overstate what the model contributes. We report it alongside the baseline and alongside the distance-matched estimate, which is the quantity that reflects the model’s own information.

^{REV}Two findings generalise past this model. The first is methodological: uncertainty evaluation on bounded bimodal targets is confounded by range compression, so that a zero-parameter heuristic can outperform genuine uncertainty estimators on the β scale while collapsing on the logit scale. Because β MAE is the standard reporting choice in methylation prediction, any model that is simply confident at extreme predictions will appear well calibrated under current practice. Scale stability provides a cheap diagnostic, and logit-space evaluation a straightforward remedy.

The second is architectural. Gated multimodal fusion improved absolute methylation prediction substantially but contributed nothing measurable to variant-effect prediction, with paired equivalence intervals bounded within ± 0.007 and no rescue through the gate. The explanation is structural rather than empirical: the context vector is identical for reference and alternate input, so it cannot create a variant effect. The implication for the field is concrete—adding richer or better-matched epigenomic context to a model of this form will not improve variant-effect prediction, whereas making context features themselves allele-dependent might. Notably, this also means that the mismatch between the breast context features used here and the blood tissue of the validation cohort is not what limits variant-effect performance.

Limitations

Methylation targets derive from cohort medians of 97 TCGA solid-tissue-normal breast samples collected adjacent to tumours, which may retain field effects or cell-composition heterogeneity [18, 31]. Epigenomic context comes from MCF-10A reference tracks rather than matched samples, and HM450 covers only a subset of genomic CpGs.

The external validation cohort is whole blood while the model’s context features are breast; results on GENOA are therefore cross-tissue transfer rather than tissue-matched validation. GENOA effect sizes are reported on a normalized-phenotype scale, so rank and sign comparisons are meaningful while magnitude calibration against them is not available.

Ancestry-stratified error on the training cohort is not adequately powered: open genetic ancestry calls for the 97 training normals yield 84 European, 4 African, and 1 South Asian participant. Cross-ancestry evidence therefore rests on the contrast between the African American GENOA cohort and European-dominant reference data, which also differ in tissue and so do not constitute a clean ancestry contrast.

Evaluation currently uses a single chromosome-holdout design; repeated chromosome-blocked splits, head-to-head comparison against CpGenie and DeepCpG, regulatory-enrichment analysis, multi-cohort evaluation on unpaired TCGA-BRCA tumours, and conformal prediction intervals are in progress and are listed in the draft-status section.

5 Data and Code Availability

Code, run instructions, and Supplementary Data are available at <https://github.com/ilikecheezits/SilentMethyl>. Primary sources are the GDC, eGTEx, GENOA, ENCODE, GENCODE, UCSC, and the HM450 annotation resource [19, 21, 24, 25, 27, 30, 32]. Harmonised GENOA variant–CpG pairs, per-seed variant scores, and the evaluation outputs underlying Tables 3 and 4 are included in the Supplementary Data.

6 Conclusion

SilentMethyl predicts CpG methylation and paired methylation differences caused by single-nucleotide variation in unseen chromosome regions. Fusion outperformed single-modality baselines for locus-level prediction. On an independent cohort of 39,657 variants differing in tissue, ancestry, and platform, predicted variant effects agreed with reported effects in direction and rank, with discrimination that survives matching on variant–CpG distance. Two accompanying findings delimit the approach: uncertainty on bounded bimodal targets must be assessed in logit space, and gated fusion cannot contribute to variant-effect prediction while its context features remain allele-invariant.

References

references.bib was not available when this draft was built, so citation keys are shown in place of formatted references. Building alongside the project bibliography restores the full list. Keys marked † are new in this revision and must be added.

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